HEATED GLASS (Continued)

- Rear window defogger switch in the instrument panel.
- Rear window defogger relay control circuity in the instrument cluster.
- Rear window defogger grid lines (all grid lines would have to be broken, or the power feed or ground wire disconnected, for the entire heated grid to be inoperative).

(4) If the EBL system operation has been verified but the rear window defogger indicator does not illuminate, check the indicator power feed fuse located in the fuse block. If the fuse is Ok, replace the rear window defogger switch (Refer to 8 - ELECTRICAL/ HEATED GLASS/REAR WINDOW DEFOGGER SWITCH - REMOVAL).

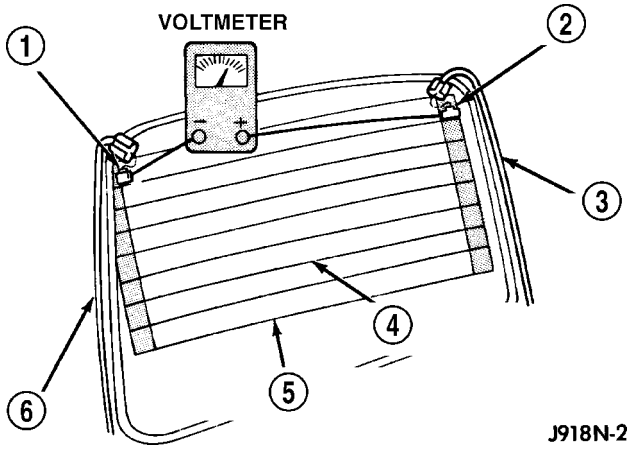


***Fig. 3 Rear Window Glass Grid Test - Typical***

1 - TERMINAL “B”
2 - TERMINAL “A”
3 - FEED WIRE
4 - MID-POINT “C” (TYPICAL)
5 - HEATED REAR WINDOW DEFOGGER GRID
6 - GROUND WIRE

# REAR WINDOW DEFOGGER RELAY

## DESCRIPTION

The rear window defogger (EBL) relay (Fig. 4) is a International Standards Organization (ISO)-type relay. Relays conforming to the ISO specifications have common physical dimensions, current capacities, terminal functions and patterns. The EBL relay is a electromechanical device that switches battery current through a fuse in the fuse block to the rear window defogger grid and to the light-emitting diode (LED) indicator of the rear window defogger switch and also to the heating elements of the outside rear view mirrors (when equipped). The EBL relay is energized when the relay coil is provided a ground path by the control circuitry within the instrument cluster.

The EBL relay is located in the power distribution center (PDC) in the engine compartment.

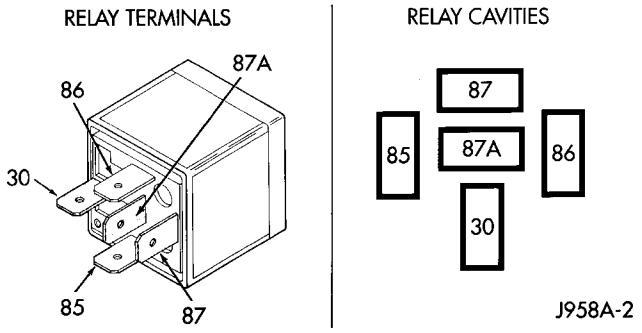


***Fig. 4 Rear Window Defogger (EBL) Relay***

30 - COMMON FEED
85 - COIL GROUND
86 - COIL BATTERY
87 - NORMALLY OPEN
87A - NORMALLY CLOSED

## OPERATION

The ISO-standard rear window defogger (EBL) relay is an electromechanical switch that uses a low current input controlled by the body control module (BCM) to control the high current output to the rear window defogger grid lines The movable, common feed relay contact is held against the fixed, normally closed relay contact by spring pressure. When the electromagnetic relay coil is energized, it draws the movable common feed relay contact away from the fixed, normally closed relay contact and, holds it against the fixed, normally open relay contact. This action allows high current to flow to the rear window defogger grid lines.

When the relay coil is de-energized, spring pressure returns the movable relay contact back against the fixed, normally closed contact point. The resistor or diode is connected in parallel with the relay coil, and helps to dissipate voltage spikes and electromagnetic interference that can be generated as the electromagnetic field of the relay coil collapses.

The EBL relay cannot be repaired and, if faulty or damaged, it must be replaced. Refer to the appropriate wiring information for diagnosis and testing of the ISO-standard relay and for complete rear window defogger (EBL) wiring diagrams.

## REMOVAL

(1) Disconnect and isolate the negative battery cable.

(2) Open the cover of the power distribution center (PDC) (Fig. 5) located in the engine compartment.